The current issue and full text archive of this journal is available on Emerald Insight at:
https://www.emerald.com/insight/0022-0418.htm

Journal of Documentation

Multivalent evaluation of digital preservation success

Stephen Abrams
Harvard Library, Harvard University, Cambridge, Massachusetts, USA

Received 30 December 2024
Revised 17 March 2025
Accepted 22 March 2025

Abstract

Purpose – Current assessment practice for digital preservation focuses on instrumental trustworthiness of processes, systems and outputs, with insufficient regard for teleological success of resulting programmatic outcomes. This paper provides a communicological formalization of digital preservation success.

Design/methodology/approach – Digital preservation is repositioned as semiotic sense-making unfolding across archival timespans and accompanying cultural as well as technical distance, and not just as persistent data management.

Findings – Inherent intersubjective contingency of any use of preserved information objects necessitates multivalent consideration of various communicative qualities, experiential affordances and artifactual characteristics for the comprehensive determination of preservation success.

Research limitations/implications – The proposed conceptual framework provides a foundation for subsequent development of evaluative best practice recommendations.

Originality/value – While digital preservation concerns span a socio-technic continuum, evaluative practice should properly emphasize the embodied experience of preservation-enabled communication. Communicological framing of preservation assessment complements the predictive reliability of programmatic trustworthiness with confirmatory assurance of experiential efficacy.

Keywords Digital preservation, Success, Intersubjectivity, Semiotics, Affordance

Paper type Conceptual paper

1. Introduction

Digital preservation is the continuation of long-standing stewardship aspirations and responsibilities of memory institutions in the digital age. The primary programmatic imperatives for preservation center on assurances regarding persistent archival integrity, authenticity, accessibility and usability of digital information resources. However, current assessment of preservation activity does not equally encompass effective measure of all these factors. Instead, it remains focused on quantifying programmatic preservation agency and technical instrumentality. The criteria embodying this technocratic perspective are most pertinent to the qualities of integrity, authenticity and accessibility. While critical to overall preservation goals, these factors should be viewed as necessary, but not fully sufficient, conditions enabling the teleological priority of subsequent experiential use. Given the inherent intersubjective contingency of any such use, development of an effective framework for evaluating the results of those engagements has remained elusive. The conceptual repositioning of the preservation enterprise as an act of human communication technically-mediated across time and space provides a new foundation for effective digital preservation assessment.

The trustworthiness of preservation programs and processes remains a widely-adopted metric for predicting positive quantitative outputs of preservation management. However, a more appropriate measure of preservation-enabled communication is its success, the relative alignment of curatorial intentions, stakeholder expectations and realized qualitative outcomes. The communicative perspective facilitates analysis of preservation activities as components of semiosis, the process of human meaning-making mediated through interplay of information-laden signs. Given the extended time horizon of preservation concern – connecting future consumers with past informative expression – preservation imperatives and evaluative criteria should focus on reception of meaning spanning disruptive cultural as well as technical distance. Thus, characterization of experiential digital preservation success is a multivalent



Journal of Documentation
Vol. 81 No. 3, 2025
pp. 747-766
© Emerald Publishing Limited
e-ISSN: 1758-7379
p-ISSN: 0022-0418
DOI 10.1108/JD-12-2024-0313

Downloaded from www.emerald.com/jd/article-pdf/81/3/747/9723661/jd-12-2024-0313.pdf by Diponegoro University user on 10 November 2026

determination, best approached through targeted consideration of various semiotic qualities and affordances embodied by preserved digital information objects and the circumstances and consequences of their preservation and use.

2. Digital preservation assessment

The need for effective socio-technical digital preservation programs arises from the centrality of information technology in contemporary life. The advent of digital culture enabled novel modes of information production, engagement and sense-making previously not possible with tangible materials. However, the inherently fragile nature of digital information leaves it susceptible to potential damage and loss through relentless socio-technical innovation, disruption and obsolescence (Dominguez Rubio and Wharton, 2020; Frank, 2020). In response, the preservation enterprise encompasses a complex of people, policies, procedures and platforms ensuring ongoing informational viability of digital artifacts constructing and documenting past and present time (Corrado and Moulaison Sandy, 2017; Traczyk, 2017; Wilson, 2017). That enterprise is most effectively provisioned in a proactive and programmatic manner (Halvarsson *et al.*, 2019; Owens, 2018) to provide unbroken continuity of stewardship (Pinnick, 2017). However, resources incumbent for success – infrastructure, staffing, funding, time – are necessarily finite. Given inevitable constraints on their availability, prudent programmatic administration dictates that they are targeted in the most productive manner possible. Effective prioritization and refinement of any programmatic activity depends on evaluative measures characterizing results in order to illuminate what works well and what could work better (Newcomer and Brass, 2016; Thorpe and Howlett, 2020). Substantive analytics also are important for addressing institutional transparency and accountability (Baer, 2017; Matthews, 2015); promoting stakeholder adoption and retention by documenting programmatic efficiency, efficacy and relevance (Jenatabadi, 2015) and garnering administrative commitment through delineation of the programmatic value proposition (Delen and Ram, 2018). All these attributes are pertinent to the digital preservation domain.

To date, however, domain assessment has focused on maturity modeling of general programmatic competency, and the specific trustworthiness of institutions and infrastructure. Maturity models are useful benchmarks for the quality of system, service and program capabilities (Tiwari and Madalli, 2021) as a proxy for operational effectiveness (Hart and Amos, 2018). Various models target the digital preservation realm, see for example (Ashley and Misic, 2019; Blumenthal *et al.*, 2016; McMeekin and Currie, 2022; Phillips *et al.*, 2013; Slade *et al.*, 2014), defining metrics regarding organizational governance, finance, policy and strategy; practitioner experience and expertise; infrastructural capacity and sustainability and procedural robustness. These are useful factors for documenting extant programmatic state, identifying operational deficiencies and strategic aspirations and planning for improvement. However, maturity assessment is essentially diagnostic, rather than predictive, in nature (Becker *et al.*, 2020), being primarily descriptive of programs and systems rather than results arising from their application. Thus, maturity is more pertinent to the instrumentality rather than the teleology of the preservation enterprise, quantifying its means but not qualifying its ends. Nevertheless, trustworthiness has long been promoted as a reliable proxy for probable, if not assured, stakeholder satisfaction.

Trustworthiness is accepted as a fundamental (Donaldson and Conway, 2015), foundational (Houghton, 2017) and core (Traczyk, 2017) digital preservation requirement. Many preservation programs invest considerable resources to assess their degree of trustworthiness; see for example (Donaldson, 2020; Elstrøm and Junge, 2014; Medina-Smith, 2021; Rodrigues *et al.*, 2014). This emphasis appears founded on an assumption of relative causality between trusted agency and positive outcomes arising from that agency. For example, Waters and Garrett (1996) posit that reliability – and, implicitly, concomitant success – of "archival processes . . . hinges on the *trustworthy* [emphasis added] operation of digital archives" (p. 11). Giaretta (2011) claims that certified trustworthiness provides an assurance

Downloaded from www.emerald.com/jd/article-pdf/81/3/747/9723661/jd-12-2024-0313.pdf by Diponegoro University user on 10 November 2026

that "digital holdings *will* [emphasis added] continue to be useable and understandable into the future" (p. 480). Bountouri *et al*. (2018) similarly state that trustworthy digital repositories "*ensure* [emphasis added] future access to digital resources" (p. 365). However, definitive or inferential claims that trustworthiness *necessarily* provides positive assurance are supported more by supposition than theory or evidence. The causal implication of trustworthiness is more limited.

Nyitray and Reijerkerk (2021) position trustworthiness as a putative indication of capability and a commitment towards – rather than achievement of – a goal. Similarly, Donaldson (2020) argues that "we do not know if TDRs [trustworthy digital repositories] preserve digital information better than their counterparts" (p. 12), and offers the qualified assertion that certification of a preservation program's trustworthiness is a determination "that it *can* [emphasis added] preserve, manage, and provide access to many types of digital materials" (p. 2). The conceptual distance between the prospect of Donaldson's "can" and the certitude of Giaretta's "will" emphasizes the critical distinction between potentiality and actuality. While trustworthiness arguably bolsters confidence in what *might* – or *is most likely to* – happen, it does not guarantee what *will* happen or confirm what actually *did* happen.

3. Evaluating managerial outputs

While trust is an important enabling condition for digital preservation, exclusive reliance on it as a reductive perspective implicitly promotes a metaphoric narrative positioning preservation as a *managerial* function concerned primarily with internal mechanics of acquiring, processing and persisting information objects. In terms of the ISO 14721 *Open Archival Information System* (OAIS) reference model (2012b), this limits the scope of preservation attention and evaluative focus to the boundaries of the OAIS itself and the artifacts preserved within it (Abrams, 2021). Bak (2016) characterizes the role of trust in preservation theory and practice as *technocratic* through its emphasis on administrative competencies and processes. While the terms "trustworthy" and "trusted" are deployed in the literature as ostensible synonyms, Bak points out that trustworthiness and trustedness entail distinct conceptual perspectives: a repository is trust*worthy* based on its internal qualities, but is trust*ed* by external agency. The former is thus susceptible to evaluation of a repository's administrative, technical and operational characteristics; the latter, by contingent and contextual perception of those characteristics in purposive use. In current practice, "trustworthy" is the form well-codified in de jure standards, for example (DIN, 2012; ISO, 2012a), as well as de facto best practice recommendations (Dale and Ambacher, 2007; Recker *et al.*, 2020). This suggests a tacit, if not intentional, choice within the preservation domain to emphasize evaluative aspects of preservation management rather than subsequent use of managed content.

The managerial concerns of digital preservation are encapsulated in broadly-adopted domain policy imperatives regarding persistent integrity, authenticity and accessibility of preserved information objects (Abrams, 2021). These correspond to artifactual qualities of those objects remaining whole and unchanged (UNESCO, 2003); being what they purport to be (Duranti *et al.*, 2022) and susceptible to discovery, request and retrieval (ISO, 2016). The OAIS Access function (ISO, 2012b, § 4.1) mediates between internal managerial concerns of the system and whatever process, agency and context lies outside, and is presented as the final obligatory OAIS goal. Access is central to many other influential definitions of the preservation field. The Digital Preservation Coalition defines preservation as "managed activities necessary to ensure continued *access* [emphasis added] to digital materials for as long as necessary" (DPC, 2015); UNESCO, as "processes aimed at ensuring the continued *accessibility* [emphasis added] of digital materials" (2019); Society of American Archivists, as "management and protection of digital information to ensure authenticity, integrity, reliability, and long-term *accessibility* [emphasis added]" (SAA, 2024) and American Library Association, as "policies, strategies and actions that ensure *access* [emphasis added] to digital content over time" (ALA, 2009).

Downloaded from www.emerald.com/jd/article-pdf/81/3/747/9723661/jd-12-2024-0313.pdf by Diponegoro University user on 10 November 2026

The latter is the ALA short definition, presumably representing the crux of preservation concern. The clarifying long definition extends the scope to encompass organizational "commitment to preserve digital content for future use" (ALA, 2009). Similarly, DPC's accompanying definition for access introduces an implicit synonymy with "ongoing usability of a digital resource" (DPC, 2015). The complementarity of accessibility and usability is fundamental to cultural heritage stewardship (Cloonan, 2007; Schüller-Zwierlein, 2015) and digital preservation (see e.g. Bountouri *et al.*, 2018; Deegan and Tanner, 2006; Strodl *et al.*, 2007). Enabling future use is the teleological justification for digital preservation (Rieger *et al.*, 2022) and involves assurances that objects remain susceptible to meaningful exploitation for desired or serendipitous purposes, or in other words, that "someone is able to do something sensible with the information it [the preserved digital object] contains" (Giaretta, 2011, p. 167). This idea of sense-making suggests that critical concerns for digital preservation encompass not just management of and access to authentic information objects, but also experiential reception of and response to those objects (Abrams, 2023a). Given the teleological priority of the latter, the proper conceptual basis for digital preservation is as an essentially communicative concern unfolding across archival timespans and accompanying cultural as well as technical distance (Abrams, 2023b).

4. Evaluating communicative outcomes

Digital preservation has been characterized as "communication with the future" (Thibodeau, 2002, p. 13); see also Brocks *et al.* (2010); Mois *et al.* (2009); Moore (2008). Use of this phrase spans a continuum from a rhetoric metaphor to fundamental conceptual foundation, but in all cases concentrating on enabling managerial instrumentalities rather than enabled communicative experiences. Thibodeau's survey (2002) posits that "the ultimate outcome of the preservation process should be authentic preserved objects" (p. 13), without consideration of subsequent engagement with those objects. Thibodeau cites Rothenberg's claim (2000) that authenticity is a contextual determination, dependent on contingent uses by socio-technically situated users, but does not explicitly factor those uses and users into the field's "ultimate outcome." Moore's (2008) theoretical frame for digital preservation is based on trustworthy processes for ensuring and validating archival authenticity and integrity as assertions about record provenance and management. Thus, these are presented as qualities inhering to preserved information objects themselves, independent of what function they might play in a communicative act embodied by the personal experiences of the human participants. While the preserved object is the medium of such an exchange, the ultimate importance of the act is some responsive change in the cognitive or affective state of the consuming actor. For example, the value of object authenticity is realized only through confident actorial reliance on that object's information content.

Mois *et al.* (2009) take an information-theoretic view of preservation-enabled communication, stating its communicative imperative as "Information that is understood today is transmitted to an unknown *system* [emphasis added] in the future where it will be interpreted and displayed" (p. 1). That interpretation is positioned as computational rather than human activity, relating to machine display as opposed to human perception and cognitive comprehension. The authors do, however, recognize the need for considerations beyond reliable transmission of encoded bitstreams in order to "guarantee that the consumer fully understands the object's content" (p. 5). These considerations include behavioral function and interpretive context. Brocks *et al.* (2010) expand the notion of context specifically to include originating creative intention and situated consuming purpose. This arises from recognition that "Production and consumption of digital content from the archive do not coincide. Preservation of digital content, especially in the long term, bridges gaps in time, space, culture, expertise, interests, and other dimensions" (p. 198). Embodied response to that content is dependent on individual lived experience and mindset, but is conditioned and constrained by internalization of productive intention, interpretive curatorial presentation, as well as

Downloaded from www.emerald.com/jd/article-pdf/81/3/747/9723661/jd-12-2024-0313.pdf by Diponegoro University user on 10 November 2026

disciplinary and cultural socialization. Thus, rather than being reductively objective, as implied by a technocratic emphasis on trustworthy management of artifactual authenticity, preservation-enabled communication occurs firmly within an intersubjective sphere (Abrams, 2023b). Intersubjectivity views the acquisition of communicative meaning as contingently arising through complex interplay or negotiated discourse between actors and contexts (Manuti *et al.*, 2012; Mascolo, 2016). Thus, preserved objects should be viewed as vehicles, not goals of preservation attention, and digital preservation outcomes should be conceived as a *process* of experiential becoming, not just a *state* of artifactual being. While inherently dependent on technical mediation to render preserved digital expression into sensible analog form, it nevertheless is ultimately an act of human communication.

Communication is concerned with "Who/Says What/In Which Channel/To Whom/With What Effect?" (Lasswell, 1948, p. 216). The final two questions implicate the communicative consumer and thus constitute concerns of information experience, the totality of "complex, multidimensional engagement with information" (Bruce *et al.*, 2014, p. 4). Communicative experience entails a future embodied response to past informative expression (Kuhlthau, 2017; Savolainen, 2015) that is meaningful when encompassing new cognitive understanding, affective impression or conative action that otherwise would not have been known, felt or performed. The degree to which responses satisfy purposive consumer context is a measure of communicative efficacy (Abrams, 2023a). While perception of programmatic trustworthiness contributes positively towards stakeholder satisfaction, that outcome is dependent on confirmation of expectations regarding use (Kim, 2012). Thus, the evaluative question for digital preservation is not merely whether enabl*ing* preservation agency is trustworthy, but rather, whether agency-enabl*ed* communicative acts were successful.

Unfortunately, methods for evaluating outcomes of digital preservation activity have not progressed significantly since 2006 when Anderson and LeFurgy (2006) highlighted the paucity of effective means to determine success. Contemporaneously, Lynch labeled success a "metric that's defied measuring" (Lee and Tibbo, 2007). Over a decade later, Poole (2016) posed the question of success metrics as an open research topic. Evans (2021) reports that capturing the "elusive" (p. 2) impact of communicative experience is possible through case studies, albeit only on a small scale and involving significant staff effort. Consequently, evaluative emphasis within the preservation domain remains focused on technocratic managerial factors bolstering programmatic dependability rather than communicative efficacy. To the extent that success is considered, it corresponds to measures of programmatic "busyness" (Creaser, 2018, p. 87) capturing what practitioners *do* but not what consumers *experience* (Brophy, 2006). In the preservation context, evaluable busyness is indicative of intermediating managerial outputs, not final communicative outcomes.

An output is a quantifiable product or service resulting from activity, while an outcome is the effect or benefit an output has on knowledge, attitudes, skills and relationships (Belcher *et al.*, 2020). Outcomes ideally lead towards impact or significant behavioral change in the wider programmatic environment (Belcher and Halliwell, 2021). Dugan and Hernon (2002) identify three evaluative perspectives pertinent to cultural heritage stewardship: that of stewarding agency, consuming agency and broader agential mission. Robertson *et al.* (2024) draw a similar distinction between conceptual spaces in which evaluative consideration can occur: sphere of control, comprising activities and results of programmatic agency; sphere of influence, encompassing those on whose behalf that agency operates and sphere of interest, reflecting lasting behavioral responses in support of goals. Within an evaluative framework, outputs are viewed as measures of the sphere of control; outcomes, the sphere of influence and impact, the sphere of interest (Gerke *et al.*, 2023). Current digital preservation evaluation focuses on outputs within the sphere of managerial control, without providing sufficient attention to outcomes within the sphere of experiential influence or the impacts within the sphere of communicative interest.

For example, the annual ALA Preservation Statistics Survey purposefully elicits measures within direct control of preservation practitioners and programs to "strike a balance between

Downloaded from www.emerald.com/jd/article-pdf/81/3/747/9723661/jd-12-2024-0313.pdf by Diponegoro University user on 10 November 2026

data that is easy for institutions to gather and data that is useful to the preservation community" (Peterson *et al.*, 2016, p. 44). The survey prompts for inventory counts, sizes and formats along with a generalized inquiry for "How confident is your institution in its ability to preserve digital content?" (ALA, 2022, p. 24). While this provides insight on managerial aspects of what preservation practitioners *control* and *do* and how they *self-reflect*, it is not directly indicative of preservation stakeholder *experience* or *impact*. Thus, it is emblematic of the fact that current evaluative measures of digital preservation are incommensurate with a conceptualization of the enterprise as a fundamentally communicative undertaking.

5. Communicological approach

Viewing digital preservation as an exercise in communication suggests its susceptibility to analysis through communicology, the study of embodied discourse considered as the semiotic expression, propagation and interpretation of meaning-laden signs within contingent systems of signification (Catt and Eicher-Catt, 2010). Saussure's (1916/1983) linguistic conception of the sign identified a dyadic relationship between the *signifiant* [signifier] and the *signifié* [signified] (Nöth, 1990). These correspond to the first two components of Morris's (1964) triadic structure of syntactics, semantics and pragmatics. This itself is a recodification of Peirce's (1902/1932) "semiotic triangle" of representamen, object and interpretant, which, like Morris's trichotomy, correspond to perceptual form, cognitive meaning and interpretive reception (Mingers and Willcocks, 2023). That reception emerges through interplay of contingent lived experience, environmental context and normative cultural codes (Catt *et al.*, 2018). Stamper's "semiotic ladder" (Mingers and Willcocks, 2023) extends the roster of semiotic dimensions for better application to digital information systems by decomposing syntactics into physics, empirics and syntactics proper, distinguishing between tangible manifestation, symbolic encoding and rhetoric expression. These concerns were further extended by Abrams (2023b) for pertinency to the digital preservation enterprise through the addition of performics (see Table 1), reflecting cognizance that behavioral performance is a necessary component of any human use of a digital resource (Becker, 2018; Wilson, 2005).

The initial rung of Stamper's ladder is designated "physical world" in reference to the physical concerns of tangible representation. It is relabeled "physics" for consistency with conventional naming of the other dimensions. Stamper's uppermost rung, "social world", reflective of aggregate effects of information within organizational settings, is similarly relabeled but also repositioned along the progression. While Stamper distinguishes between the individual response of the pragmatic from the collective consequence of the social, the communicological underpinnings of this analysis emphasize the interpretive perspective of the individual, although still recognizing the inherent intersubjective – and thus, socially contingent – nature for that interpretation. That contextual interpretation is recast as "plaistics" – from the Greek πλαίσιο [*plaisio*], "frame" – as an intermediating rung between performic behavior and pragmatic understanding (Abrams, 2023a). In terms of preservation imperatives, the pragmatic reception of an individual consumer, that is, the idio-pragmatic, is a more proximate concern than subsequent effects within a wider social milieu, that is, the socio-pragmatic.

Each of the seven semiotic dimensions corresponds to a core preservable – and evaluable – quality (see Table 2). They represent an expansion of the four primary evaluative principles – integrity, authenticity, accessibility, usability – previously determined as tacit underpinnings of preservation discourse (Abrams, 2021). The expanded roster offers a more comprehensive and granular perspective on the pertinent characteristics of preserved digital information. The physical integrity of the reified manifestation of an information object indicates that it is whole and unchanged from its canonical state (Ross, 2006); the empiric validity of its representational encoding, that it conforms to authoritative requirements of its declared format (Shala and Shala, 2016); the syntactic accuracy of its rhetorical expression, that it faithfully represents its underlying meaning (Donaldson and Conway, 2015); the semantic

Downloaded from www.emerald.com/jd/article-pdf/81/3/747/9723661/jd-12-2024-0313.pdf by Diponegoro University user on 10 November 2026

Table 1. Communicologically-based preservation ladder

Stamper's semiotic ladder	Physical world	Empirics	Syntactics	Semantics			Pragmatics	
Communicological preservation ladder	Physics	Empirics	Syntactics	Semantics	Performics	Plaistics	Idio-Pragmatics	Socio-

Source(s): Authors' own work

Downloaded from www.emerald.com/jd/article-pdf/81/3/747/9723661/jd-12-2024-0313.pdf by Diponegoro University user on 10 November 2026

Table 2. Communicological foundations of preservation evaluation

Perspective	Managerial				Communicological		
Concern	Information object				Info performance	Information experience	
Scope	Constitutive isness		Denotative ofness			Connotative aboutness	
Abstraction	Carrier	Message			Channel	Environment	Mind
Dimension	Physics	Empirics	Syntactics	Semantics	Performics	Plaistics	Pragmatics
Role	Reificatory	Representational	Rhetorical	Existential	Presentational	Relational	Experiential
Purpose	Manifestation	Encoding	Expression	Meaning	Behavior	Context	Understanding
Quality	Integrity	Validity	Accuracy	Reliability	Availability	Relevancy	Efficacy
	Authenticity				Accessibility	Legitimacy	
Materiality	Ontological				Epistemological	Phenomenological	
Basis	Immanent		Emergent				
Measure	Resulting outputs		Impactful outcomes				
Warrant	Objective		Intersubjective				
Purpose	Instrumental means				Teleological ends		
Evaluation	Enabling trustworthiness				Enabled success		
	Necessity →		→		→	Sufficiency	

Source(s): Author's own work

reliability of that existential meaning, that it is factually truthful (Kastenhofer, 2015); the performic availability of its presentational behavior, that it can be meaningfully engaged with (Burda and Teuteberg, 2013); the plaistic relevance of its relational context, that it supports information-seeking expectations and shapes final interpretive sense-making (Beaudoin, 2012) and the pragmatic efficacy of its experiential understanding, that it satisfies a premeditated or serendipitous purpose (Abrams, 2023a).

The semiotic dimensions and their associated roles, purposes and qualities are categorized by the scope of their informative characterization. In cataloging theory, *isness*, *ofness* and *aboutness* correspond to capacities illuminating what a cataloged thing is, what it overtly represents and what it is interpretively concerned with (Golub *et al.*, 2022; Hjørland, 2023). In the digital preservation context, isness encompasses the factual composition of an object's informative constitution; ofness, the overt and widely accepted denotative meaning of the object and aboutness, the nuanced contingency of its interpretive connotation (Mingers and Willcocks, 2023). For example, consider the case of this article under preservation stewardship. Its isness relates physically to being a bitstream on some storage medium, and empirically to being a valid PDF-encoded file. Its ofness relates syntactically to an accurate text conforming to rhetorical conventions for scholarly research, semantically to a reliable derivation of a novel multivalent model for evaluating digital preservation success, and performically to an accessible document readable via technical rendering, page navigation, textual search, etc. Finally, the article's aboutness relates plaistically to a series of contextualizing publications on similar topics by its author as well as relevant discourse within the digital preservation community, and pragmatically to the legitimate reception and acceptance of – or argument with – a critique of prior modes of domain assessment and recommendation for future evaluative practices.

The imperative qualities implicitly define criteria for subsequent evaluation of preservation success based on the central question, Has a given communicative quality been persisted? The resulting outputs associated with preservation of physical and empiric roles and purposes are inherent *within* the fabric of the preserved information object, while subsequent impactful outcomes regarding semantics, performics, plaistics and pragmatics emerge dynamically *from* the contingent interplay of object, performance and experience. The nominal point of division is aligned with the syntactic dimension, reflecting postmodern archival recognition that authenticity and reliability are not fixed, but rather, free-floating and contextually-dependent qualities (MacNeil and Mak, 2007): there is "not one narrative . . . but many narratives, many stories, serving many purposes for many audiences, across time and space" (Cook, 2001, p. 7). The evidentiary warrant pertinent to assessment of preservable qualities ranges between the

Downloaded from www.emerald.com/jd/article-pdf/81/3/747/9723661/jd-12-2024-0313.pdf by Diponegoro University user on 10 November 2026

immanently objective and emergently intersubjective, broadly aligned with focused concerns of complementary managerial and communicative perspectives.

6. Communicological imperatives

Integrity of manifestation, validity of encoding, accuracy of expression and reliability of meaning can be subsumed under an encompassing quality of artifactual *authenticity*, the attribute of an object being what it purports to be (Duranti *et al.*, 2022). While reliability is the most teleologically prominent in this progression, and would otherwise provide a more appropriate heading, authenticity is used to conform to common usage in digital preservation discourse, where reliability is referenced primarily in regard to systems and programs rather than objects; see for example (Altman and Landau, 2024; Constantopoulu *et al.*, 2005; Rieger *et al.*, 2022). Authenticity also is an appropriate meta-concept as it implicates the inherent dynamic and temporal nature of preservation. Reliability is characteristic of the information content of a preserved object dependent on pertinent circumstances of its production, and thus the primary *point-in-time* responsibility of creative and acquisitional agency. Authenticity, on the other hand, is characteristic of the persistent stewardship of the content's expressive form, which needs to be proactively protected, assessed, possibly reestablished and maintained *over time* (Duranti and Blanchette, 2004), and thus the primary ongoing responsibility of preservation agency.

Similarly, relevancy of context and efficacy of understanding fall underneath the inclusive quality of experiential *legitimacy*, the intersubjectively-contingent sense of satisfaction with an engagement with the object. The quality of availability is recast as *accessibility* to take advantage of that term's contemporary as well as traditional meaning. Traditionally, accessibility refers to an information object being known to exist, findable, requestable and retrievable with respect to technical capacity, legal status and curatorial policy (SAA, 2020). More recently, accessibility has come to mean sensitivity and responsiveness to the widest spectrum of physical and cognitive capabilities (Botelho, 2021). This encompasses facilitating information engagements at a time and place and in a manner satisfying the consumer's purposive perceptual and intellectual needs and preferences. In this sense, accessibility explicitly entails the notion of contextual suitability that is central to the communicative foundation for preservation success.

The preservation qualities of authenticity, accessibility and legitimacy align with contemporary LIS discourse regarding the nature of digital materiality, grounded in concern for ontological essence; epistemological presentation, through which essence is perceived and phenomenological reception of and response to that essence (Sundqvist, 2021). This decouples the historical synonymy of materiality with *physical* substance, emphasizing instead the significance of *informational* instantiation and *experiential* consequence (Leonardi, 2010). With these meta-concepts in hand, the digital preservation enterprise coalesces into three high-level imperatives needing persistent attention, analysis, action and evaluation:

(1) Assuring authentic information objects

(2) Provisioning accessible information performances

(3) Enabling legitimate information experiences

The nominal *object* → *performance* → *experience* progression in substantive focus represents a stepwise expansion of preservation concern to embrace the full communicative breadth of information carrier, behavior and understanding. It also implies a concomitant expansion in applicability from generic universal awareness of the extrinsic objective carrier to specific individual awareness of intrinsic intersubjective experience. Across the qualitative *authentic* → *accessible* → *legitimate* progression, authenticity straddles the objective/intersubjective divide, while accessibility and legitimacy occupy the intersubjective end of the spectrum. Determinations of authenticity may not be fully objective, but they are more firmly-rooted in

Downloaded from www.emerald.com/jd/article-pdf/81/3/747/9723661/jd-12-2024-0313.pdf by Diponegoro University user on 10 November 2026

the artifactual fabric, extrinsic characterization and managerial environment of the communicative carrier (Duranti *et al.*, 2022), and thus more likely to be commonly acknowledged and shared. Evaluable claims of accessibility and legitimacy, on the other hand, are inherently particular to time, place, person and purpose.

The varying degrees of modal commitment underlying the *assuring* → *provisioning* → *enabling* progression correspond to programmatic obligations (i.e. conditions the program *must* assure), expectations (affordances it *should* provide) and intentions (opportunities it *could* enable). These positions encompass both deontic and epistemic modality, the former indicating degree of requisite necessity regarding a particular state of affairs, and the latter, probability that that state is subsequently realized (Nuyts, 2006). Intersubjective contingency of internal experience is inherently more slippery to characterize – and thus, confidently assure and measure – relative to intermediating epistemic behaviors or artifactual externalities of preserved information objects. Thus, the progressive *expansion* in intersubjective pertinence across the three imperative preservation goals is necessarily accompanied by a *relaxation* of imperative commitment and confidence. Preservation agency can justifiably assert the highest level of confident commitment regarding the substantive aspects of the enterprise under its direct control: the authentic artifactual state of the information objects under proactive curation. While provision of behavioral modalities revealing the information content of those objects also is a programmatic responsibility, accessibility of resulting performances only can be evaluated with respect to the needs of external consuming agency. Similarly, contingent experiential reception of those performances is largely outside of direct programmatic control, so their ultimate success is realistically subject to lower degrees of programmatic commitment and evaluable confidence.

7. Preservable properties, characteristics, affordances and qualities

Consideration of digital preservation success is dependent on understanding what aspects of a digital object could, should, must and have been preserved (Becker, 2018). Historically, the tripartite preservation goals of authenticity, accessibility and legitimacy have focused on the significant properties (Hockx-Yu and Knight, 2008), significant characteristics (Dappert and Farquhar, 2009) or significant knowledge (Boutard and Guasavino, 2012) of to-be-preserved digital objects. These attributes constitute the fabric of an object's ontological, epistemological and phenomenological being – whether immanent or emergent – and support its creative, curatorial and exploitative purpose and value. Significant attributes have been identified for a variety of content genres and types (Knight and Pennock, 2009; van Veenendaal *et al.*, 2018). As Yeo (2010) notes, however, significance is not a universal. Instead, significance is determined differentially by all stakeholders – content producers, curators, service providers and consumers – implicated in the preservation enterprise (Dappert and Farquhar, 2009). The concept of significance is also problematic with regard to its reductive dichotomous nature. Any assertion that some subset of preservable properties is *the* significant set implicitly relegates all others to insignificance. However, given a large enough stakeholder community and long enough time horizon, "all properties are significant to someone" (Bak, 2024, p. 862). Thus, it is more appropriate to refer to *preservable* properties rather than significant ones, and to evaluate all properties across a continuous and contextually-relative, rather than binary and universally-absolute, scale of importance.

Given this inherent subjectivity, it is useful to think not merely in terms of artifactual-centric attributes, but also experiential-centric affordances (Hedstrom and Lee, 2002). An affordance is a functional power enabling opportunities for consuming agents to exploit affording objects in a purposive, albeit contingent, context (Cheikh-Ammar, 2018). Supplementing an evaluable framework of preservable attributes with affordances emphasizes the primacy of consumer agency regarding preservation success. While properties pertain to preserved objects independent of use, affordances are pertinent to the exploitation of those properties through consumer use. Thus, affordancial outcomes bolster stewardship claims of assurance regarding preservation's seven semiotic qualities.

Downloaded from www.emerald.com/jd/article-pdf/81/3/747/9723661/jd-12-2024-0313.pdf by Diponegoro University user on 10 November 2026

Affordances and qualities are an extension of Dappert and Farquhar's (2009) hierarchy of preservable properties and characteristics (see Table 3). A preservable *property* is a characterizing value-space of a characterizable entity, while a *characteristic* is an instantiated property/value pair. Similarly, a preservable *affordance* is an informative function supported by an informing characteristic, while a *quality* is an exercised characteristic/function pair. For example, the file-level MD5 property and its value constitute a file's message digest characteristic. This characteristic supports the functional opportunity for verifying the digest value; together, they constitute the fixity affordance. When that function is exercised, and verifies adherence to a file's canonical bit-level state, it justifies experiential reliance on the quality of file-level integrity.

Note that MD5-based message digests are cryptographically insecure (Wang and Yu, 2005). While MD5-based fixity is appropriate for detecting and correcting inadvertent or random technical failure, it is insufficient to protect against malicious attack, particularly in instances where evidentiary or diplomatic integrity, and thus authenticity, is paramount due to policy, regulatory or statutory obligations. To support the fullest range of contextual priorities for cryptographic assurance, the fixity affordance should encompass other characteristics based on the SHA-2 or SHA-3 digest families. Thus, choice of the characteristic basis for any given quality is inherently contextually dependent.

In general, a preservable quality may cohere out of the confluence of multiple affordances, which in turn may depend upon multiple characteristics. For example, the quality of empiric validity relies on an affordancial function conforming adherence of format-specific characteristics to the requirements and prohibitions of that format's authoritative specification. For a complex format, such as PDF, the characteristic set can be substantial. The Arlington machine-readable definition of the PDF 2.0 document object model (Wyatt, 2021) identifies over 600 object types and 3,900 properties that must or can occur, many of which have specific contextual constraints governing their empiric, syntactic or semantic usage. In this case, the validity quality and its confirmatory affordance are sustained only through comprehensive examination of all implicated characteristics.

8. Multivalent evaluation model

The complexity of dealing with a multiplicity of preservable qualities, affordances and characteristics is compounded by the intersubjective nature of preservation-enabled communication. Individual or sets of preservable attributes for a given object may be

Table 3. Hierarchy of preservable attributes

Definition	Preservable property *has* Characteristic value	Preservable characteristic *supports* Affordancial function	Preservable affordance *enables* Qualitative experience	Preservable quality
Example	MD5 = d41d8cd9 … 7e	Message digest *supports* Verification of physical state	Fixity *enables* Reliance on physical state	Integrity

Source(s): Authors' own work

Downloaded from www.emerald.com/jd/article-pdf/81/3/747/9723661/jd-12-2024-0313.pdf by Diponegoro University user on 10 November 2026

exploited by consuming agents in a variety of contexts with a variety of degrees of importance for a variety of purposes. For example, this article legitimately can be considered as either a domain-specific expression of an intellectual argument or a domain-agnostic documentary artifact. LIS scholars interested in digital preservation would generally prioritize the former, consuming the article as a semantically-signifying text stream independent of layout, pagination, font and other incidental visual artifacts of design. On the other hand, scholars approaching the article from the perspective of descriptive, analytic or textual bibliography (Hjørland, 2024) would tend to emphasize the latter, finding primary significance precisely in those syntactic details. Each case is equally valid and potentially unique with regard to preservable attributes most pertinent to the underlying contextual purpose. What needs to be preserved to support the former may be radically different from the latter, implying variant criteria for evaluation of the success. The overall programmatic success of the preservation of this article is dependent on satisfactory outcomes for both, and other, use cases. The complexity and diversity of known, anticipated or even unexpected agential purposes suggest a need to approach evaluation in an explicitly multivalent manner.

Multivalent evaluation of programmatic digital preservation success can be modeled in terms of objects, consumers, contexts, qualities, affordances and characteristics, formally represented as:

$$\mathbb{S} = \sum_{o \in O} \sum_{u \in U_o} \sum_{x \in X_{o,u}} \sum_{q \in Q_{o,u,x}} \sum_{a \in A_{o,u,x,q}} \sum_{p \in P_{o,u,x,q,a}} \frac{\omega_{o,u,x,q,a,p} \cdot s_{o,u,x,q,a,p}}{k_{o,u,x,q,a}} \tag{1}$$

$\mathbb{S}$ is the aggregate multivalent success score for a preservation program; O is the set of digital objects under programmatic stewardship; U_o is the set of consumers engaging with object $o \in \mathrm{O}$; $X_{o,u}$ is the set of contingent contexts for $u \in U_o$; $Q_{o,u,x}$ is the set of preservable qualities implicated in context $x \in X_{o,u}$; $A_{o,u,x,q}$ is the set of affordances enabling quality $q \in Q_{o,u,x}$ and $P_{o,u,x,q,a}$ is the set of characteristics supporting affordance $a \in A_{o,u,x,q}$. Weighting factor $\omega_{o,u,x,q,a,p}$ indicates the relative importance ascribed or imputed to engagement with object o by consumer u in context x regarding quality q enabled by affordance a supported by characteristic p. Individual success score $s_{o,u,x,q,a,p}$ indicates the level of contingent satisfaction with that engagement. Weights and scores are defined in the inclusive real unit range $[0.0, 1.0]$. Scaling factor $k_{o,u,x,q,a} = |O| \cdot |U_o| \cdot |X_{o,u}| \cdot |Q_{o,u,x}| \cdot |A_{o,u,x,q}| \cdot |C_{o,u,x,q,a}|$ normalizes $\mathbb{S}$ to that same unit range.

While susceptible to seemingly precise mathematical formulation, it is not realistic for success to be evaluated as an exact quantitative measure. Equation (1) is better viewed as a useful conceptual framework for qualitative assessment. Rather than assuming an impracticable determination of fine continuous gradations of weights and scores, tractable but still informative evaluation is possible through a discrete set of categories. Weighting of importance could be specified on a five-point scale: $\omega \in \{0.0, 0.25, 0.5, 0.75, 1.0\} \equiv \{$*negligible*, *minor*, *nominal*, *strong*, *critical*$\}$, with success similarly scored as {*minimal*, *weak*, *nominal*, *reasonable*, *maximal*}. This coarser granularity acknowledges the inherent difficulty in asserting pertinent distinctions in satisfaction arising from intersubjective experience. It is important that any final reported success does not imply a greater degree of precision than is defensible.

Even with discretization of measure and report, the per-object × user × context × quality × affordance × characteristic combinatorics of Equation (1) leaves its application prohibitively complex. Further simplification is possible by shifting evaluative focus from the individual to the aggregate. OAIS designated communities are defined in terms of their assumed common knowledge base for understanding – and making effective use of – preserved content (ISO, 2012b), as well as common information seeking behavior; see for example (Frank and Rothfritz, 2023; Moles, 2021). Thus, DCs are aggregate proxies for consumers sharing contexts of use; qualitative, affordancial and characteristic preferences; and evaluable weights. Community

Downloaded from www.emerald.com/jd/article-pdf/81/3/747/9723661/jd-12-2024-0313.pdf by Diponegoro University user on 10 November 2026

$c \in C_o$ is a tuple $(X_{o,c}, Q_{o,c,x}, A_{o,c,x,q}, P_{o,c,x,q,a}, \omega_{o,c,x,q,a,p})$, where C_o is the set of all DCs interested in object o; $X_{o,c}$ is c's engagement contexts; $Q_{o,c,x}$ is x's implicated qualities; $A_{o,c,x,q}$ is q's enabling affordances; $P_{o,c,x,q,a}$ is a's supporting characteristics and $\omega_{o,c,x,q,a,p}$ is c's characteristic weighting factor. Substituting C_o for U_o in Equation (1) yields:

$$\mathbb{S} = \sum_{o \in O} \sum_{c \in C_o} \sum_{x \in X_{o,c}} \sum_{q \in Q_{o,c,x}} \sum_{a \in A_{o,c,x,q}} \sum_{p \in P_{o,c,x,q,a}} \frac{\omega_{o,c,x,q,a,p} \cdot s_{o,c,x,q,a,p}}{k_{o,c,x,q,a}} \tag{2}$$

with reasonable assumption that $|X_{o,c}| \leq |X_{o,u}|$, $|Q_{o,c,x}| \leq |Q_{o,u,x}|$, $|A_{o,c,x,q}| \leq |A_{o,u,x,q}|$, $|P_{o,c,x,q,a}| \leq |P_{o,u,x,q,a}|$, and, most importantly, $|C_o| \ll |U_o|$. Thus, the evaluative complexity implied by Equation (2)'s underlying Cartesian product is of order $\mathcal{O}(k_{o,c,x,q,a}) \ll \mathcal{O}(k_{o,u,x,q,a})$, significantly increasing its tractability by minimizing unique points of measurement.

To illustrate the model's application, consider an image object captured through digital photography of a museum painting. One (or more) consumers, or DCs, could intend use of the object in two contexts: as an approximate surrogate, for example, as an indicative thumbnail in a catalog record; or as an exacting facsimile of the original to produce the highest quality fine art reproduction for exhibit or sale. Both cases implicate the quality of expressive syntactic accuracy, but at varying degrees of sensitivity: in the former case, "good enough" visual accuracy to support effective patron discovery; in the latter, colorimetric and dimensional equivalence with the original. The latter are enabled by two affordances: colorimetric fidelity and spatial fidelity, each supported by specific characteristics, properties and values ensuring that reproduced size and color palette conform to the original. The two use cases can reasonably assert different weighting factors for these preservable attributes (see Table 4).

For purposes of surrogacy, colorimetric and spatial fidelity are minor or nominal considerations. While potentially desirable, they are not necessary in the same way they are critical for the facsimile use case. By explicitly highlighting these and similar nuances, the multivalent model provides insights useful for preservation decision-making, for example, selection of appropriate formats for object capture or production, ingest normalization and subsequent interventions such as format migration. A surrogate image object can be represented, and preserved, without carrying vital characteristic information internal to its structure or as external metadata, while a facsimile object must be embodied with or accompanied by that information. Normalizing or migrating an object from a format cognizant of colorspace and spatial metrics to one that isn't would represent a preservation failure for the facsimile use case, but some nominal degree of success for the surrogate case. In an ideal preservation environment *all* possible attributes would be maintained. In practice, however, policy and resource constraints always lead to a need to balance effort and outcome.

Table 4. Example qualities, affordances and characteristics

Preservable attributes					Contextual weight	
Quality	Affordance	Characteristic	Property	Value	Surrogate	Facsimile
Accuracy	Colorimetric fidelity	Colorspace	Color profile	opRGB	Minor	Critical
	Spatial fidelity	Sampling plane	Plane	Source	Minor	Critical
		Sampling frequency	Unit	PPI	Minor	Critical
			X	*x*	Minor	Critical
			Y	*y*	Minor	Critical
		Image Size	Height	*h*	Nominal	Critical
			Width	*w*	Nominal	Critical

Source(s): Author's own work

Downloaded from www.emerald.com/jd/article-pdf/81/3/747/9723661/jd-12-2024-0313.pdf by Diponegoro University user on 10 November 2026

The multivalent model offers new perspective supporting prudent, but responsible, preservation planning and action.

9. Conclusion

If, as suggested above, the multivalent evaluation model is not intended as a quantitatively-rigorous means for characterization of digital preservation success, what value does it present to the preservation community? Foremost, it promotes a conceptual perspective of digital preservation as a communicative enterprise, extending the field's historical emphasis on assurances regarding persistence of authentic information objects with complementary concerns for accessible information performances and legitimate information experiences. By incorporating recognition of the teleological preeminence of these embodied human experiences, the model repositions preservation activity within an inherently intersubjective sphere of contextually-contingent intention, expectation and assessment. It also expands the scope of assessment beyond its current focus on enabling trustworthiness of managerial programs and systems by establishing an alternative evaluative benchmark of the successful enabled outcomes of those programs and systems. The model's communicological basis of usage-centric semiotic roles, purposes, qualities and affordances, and not just their supporting artifactual characteristics, implicitly reinforces the primacy of consumer satisfaction in the determination of that success. By offering provision for incorporation of criteria spanning managerial and communicative concerns, the model provides a uniform framework for combining the predictive power of programmatic trustworthiness with the confirmatory assurance of experiential efficacy. Finally, the model offers an approach to the long-problematic concepts of significant properties and designated communities, recasting them as inclusionary rather than exclusionary proxies contextually-representative of the universe of contingently-diverse stakeholder needs and purposes.

The work presented here provides a foundational framework for thinking about evaluation of digital preservation success. The model is the conceptual springboard for subsequent investigation to further delineate and seek consensus concerning the preservable qualities, affordances and characteristics underpinning preservation success. The full value of the model awaits future realization as a pedagogic instrument for insight into the preservation domain and an operational means of characterizing the programmatic efficacy of long-term digital stewardship.

Digital preservation theorists and practitioners occupy a position of significance regarding grand challenges and opportunities attendant to effective stewardship of the essential, but inherently fragile, digital legacy of past and contemporary life for future reference and insight. Without recourse to pertinent measures of preservation outcomes, advocacy for sustained institutional commitment, operational prioritization and allocation of finite programmatic resources and ethical accountability to stakeholders is difficult. Metrical pertinence requires consideration of the intrinsic communicative nature of the preservation enterprise. Cognizance of the semiotic breadth and nuance of communicative concerns, and the innate intersubjective contingency obtaining to preservation as a semiotic process, suggests a multivalent approach to evaluation of programmatic success. The application of this approach should provide more reliable and productive evidence for purposes of planning, implementing, reflecting on and refining the means for achieving desirable digital preservation goals and fulfilling diverse stakeholder needs and aspirations.

References

Abrams, S. (2021), "Tacit attitudinal principles for evaluating digital preservation success", *Archival Science*, Vol. 21 No. 3, pp. 295-315, doi: 10.1007/s10502-021-09360-5.

Abrams, S. (2023a), *A Communicological Critque of Evaluative Norms for Digital Preservation Success*, Dissertation, Queensland University of Technology, Brisbane, doi: 10.5204/thesis.eprints.238194.

Downloaded from www.emerald.com/jd/article-pdf/81/3/747/9723661/jd-12-2024-0313.pdf by Diponegoro University user on 10 November 2026

Abrams, S. (2023b), "Rethinking digital preservation: conceptual foundations", *paper presented at 19th International Conference on Digital Preservation*, Urbana, available at: https://www.ideals.illinois.edu/items/128305

ALA (2009), *Definitions of Digital Preservation*, American Library Association, available at: http://www.ala.org/alcts/resources/preserv/defdigpres0408 (accessed 7 January 2024).

ALA (2022), *FY2022 Preservation Statistics Survey*, American Library Association, Chicago, available at: https://www.ala.org/sites/default/files/core/content/resources-tools/FY2022%20Preservation%20Statistics%20Survey%20Questions.pdf

Altman, M. and Landau, R. (2024), "Selecting efficient and reliable preservation strategies: modeling long-term information integrity using largescale hierarchical discrete event simulation", *International Journal of Digital Curation*, Vol. 18 No. 1, pp. 1-24, doi: 10.2218/ijdc.v18i1.743.

Anderson, M. and LeFurgy, B. (2006), "The archive ingest and handling test: implications of diverse content and diverse repository practices", *paper presented at Digital Curation and Trusted Repositories: Seeking Success Workshop*, Chapel Hill, available at: https://ils.unc.edu/tibbo/JCDL2006/Lefurgy_jcdl_paper_3_wgl%20_2_.pdf

Ashley, L. and Misic, M. (2019), "Digital preservation capability maturity model (DPCMM): genesis and practical uses", in Katuu, S. (Ed.), *Diverse Applications and Transferability of Maturity Models*, IGI Global, pp. 152-167, doi: 10.4018/978-1-5225-7080-6.ch006.

Baer, L.L. (2017), "Connecting the dots: accountability, assessment, analytics, and accreditation", *Planning for Higher Education*, Vol. 46 No. 1, pp. 1-16, available at: http://search.proquest.com/scholarly-journals/connecting-dots-accountability-assessment/docview/2008824562/se-2

Bak, G. (2016), "Trusted by whom? TDRs, standards culture and the nature of trust", *Archival Science*, Vol. 16 No. 4, pp. 373-402, doi: 10.1007/s10502-015-9257-1.

Bak, G. (2024), "Digital provenance", *Archival Science*, Vol. 24 No. 4, pp. 847-869, doi: 10.1007/s10502-024-09462-w.

Beaudoin, J.E. (2012), "Context and its role in the digital preservation of cultural objects", *D-lib Magazine*, Vol. 18 Nos 11/12, pp. 11-12, doi: 10.1045/november2012-beaudoin1.

Becker, C. (2018), "Metaphors we work by: reframing digital objects, significant properties, and the design of digital preservation systems", *Archivaria*, Vol. 85, pp. 6-36, available at: http://muse.jhu.edu/article/694332

Becker, C., Maemura, E. and Moles, N. (2020), "The design and use of assessment frameworks in digital curation", *Journal of the Association for Information Science and Technology*, Vol. 71 No. 7, pp. 55-68, doi: 10.1002/asi.24209.

Belcher, B. and Halliwell, J. (2021), "Conceptualizing the elements of research impact: towards semantic standards", *Humanities and Social Sciences Communication*, Vol. 8 No. 183, 183, doi: 10.1057/s41599-021-00854-2.

Belcher, B.M., Davel, R. and Claus, R. (2020), "A refined method for theory-based evaluation of the societal impacts of research", *MethodsX*, Vol. 7, 100788, pp. 1-20, doi: 10.1016/j.mex.2020.100788.

Blumenthal, K.-R., Griesinger, P., Kim, J., Peltzman, S. and Steeves, V. (2016), "What makes a digital steward: a competency profile based on the National Digital Stewardship Residencies", *paper presented at 13th International Conference on Digital Preservation*, Bern, available at: https://osf.io/zndwq.

Botelho, F.H.F. (2021), "Accessibility to digital technology: virtual barriers, real opportunities", *Assistive Technology*, Vol. 33 No. sup1, pp. 27-34, doi: 10.1080/10400435.2021.1945705.

Bountouri, L., Gratz, P. and Sanmartin, F. (2018), "Digital preservation: how to be trustworthy", in Ioannides, M. (Ed.), *Digital Cultural Heritage*, Springer, pp. 364-374, doi: 10.1007/978-3-319-75826-8_29.

Boutard, G. and Guasavino, C. (2012), "Archiving electroacoustic and mixed music: significant knowledge involved in the creative process of works with spatialisation", *Journal of Documentation*, Vol. 68 No. 6, pp. 749-771, doi: 10.1108/00220411211277028.

Downloaded from www.emerald.com/jd/article-pdf/81/3/747/9723661/jd-12-2024-0313.pdf by Diponegoro University user on 10 November 2026

Brocks, H., Kranstedt, A., Jaschke, G. and Hemmje, M. (2010), "Modeling context for digital preservation", in Szczerbicki, E. and Nguyen, N.T. (Eds), *Smart Information and Knowledge Management: Advances, Challenges, and Critical Issues*, Springer, pp. 197-226, doi: 10.1007/978-3-642-04584-4.

Brophy, P. (2006), *Measuring Library Performance: Principles and Techniques*, Facet, London.

Bruce, C., Davis, K., Hughes, H., Partridge, H. and Stoodley, I. (2014), "Information experience: contemporary perspectives", Bruce, C., Davis, K., Hughes, H., Partridge, H. and Stoodley, I. (Eds), *Information Experience: Approaches to Theory and Practice*, Emerald, pp. 3-15, doi: 10.1108/S1876-056220149.

Burda, D. and Teuteberg, F. (2013), "Sustaining accessibility of information through digital preservation: a literature review", *Journal of Information Science*, Vol. 39 No. 4, pp. 442-458, doi: 10.1177/0165551513480107.

Catt, I.E. and Eicher-Catt, D. (2010), "Communicology: a reflexive human science", in Catt, I.E. and Eicher-Catt, D. (Eds), *Communicology: the New Science of Embodied Discourse*, Fairleigh Dickinson University Press, pp. 15-29.

Catt, I.E., Klyukanov, I.E. and Smith, A.R. (2018), "Communicology: what's in a name?", Smith, A.R., Catt, I.E. and Klyukanov, I.E. (Eds), *Communicology for the Human Sciences: Lanigan and the Philosophy of Communication*, Peter Lang, pp. 1-20.

Cheikh-Ammar, M. (2018), "The IT artifact and its spirit: a nexus of human values, affordances, symbolic expressions, and IT features", *European Journal of Information Systems*, Vol. 27 No. 3, pp. 278-294, doi: 10.1080/0960085X.2018.1436025.

Cloonan, M.l. V. (2007), "The paradox of preservation", *Library Trends*, Vol. 56 No. 1, pp. 133-147, doi: 10.1353/lib.2007.0047.

Constantopoulu, P., Doerr, M. and Petraki, M. (2005), "Reliability modelling for long term digital preservation", *Paper Presented at Digital Repositories: Interoperability and Common Services, 9th DELOS Network of Excellence*, Heraklion.

Cook, T. (2001), "Archival science and postmodernism: new formulations for old concepts", *Archival Science*, Vol. 1 No. 1, pp. 22-24, doi: 10.1007/BF02435636.

Corrado, E.M. and Moulaison Sandy, H. (2017), *Digital Preservation for Libraries, Archives, and Museums*, Rowman & Litlefield, Lanham.

Creaser, C. (2018), "Assessing the impact of libraries – the role of ISO 16439", *Information and Learning Science*, Vol. 119 Nos 1/2, pp. 87-93, doi: 10.1108/ILS-05-2017-0037.

Dale, R.L. and Ambacher, B. (2007), *Trustworthy Repositories Audit & Certification: Criteria and Checklist*, Libraries, C. f. R, available at: https://www.crl.edu/sites/default/files/d6/attachments/pages/trac_0.pdf

Dappert, A. and Farquhar, A. (2009), "Significance is in the eye of the stakeholder", *Paper Presented At 13th European Conference On Research And Advanced Technology For Digital Libraries*, pp. 297-308, doi: 10.1007/978-3-642-04346-8_29, available at: http://www.planets-project.eu/docs/papers/Dappert_Significant_Characteristics_ECDL2009.pdf

Deegan, M. and Tanner, S. (2006), "Key issues in digital preservation", in Deegan, M. and Tanner, S. (Eds), *Digital Preservation*, Facet, pp. 1-31.

Delen, D. and Ram, S. (2018), "Research challenges and opportunities in business analytics", *Journal of Business Analytics*, Vol. 1 No. 1, pp. 2-12, doi: 10.1080/2573234X.2018.1507324.

DIN (2012), "Information and documentation – criteria for trustworthy digital archives (DIN 31644)", doi: 10.31030/1854419.

Dominguez Rubio, F. and Wharton, G. (2020), "The work of art in the age of digital fragility", *Public Culture*, Vol. 32 No. 1, pp. 215-245, doi: 10.1215/08992363-7816365.

Donaldson, D.R. (2020), "Certification information on trustworthy digital repository websites: a content analysis", *PLoS One*, Vol. 15 No. 12, e0242525, doi: 10.1371/journal.pone.0242525.

Donaldson, D.R. and Conway, P. (2015), "User conceptions of trustworthiness for digital archival documents", *Journal of the Association for Information Science and Technology*, Vol. 66 No. 12, pp. 2427-2444, doi: 10.1002/asi.23330.

Downloaded from www.emerald.com/jd/article-pdf/81/3/747/9723661/jd-12-2024-0313.pdf by Diponegoro University user on 10 November 2026

DPC (2015), "Digital preservation handbook", in Beagrie, N. (Ed.), *Digital Preservation Coalition*, 2nd ed., Glasgow, available at: https://www.dpconline.org/handbook

Dugan, R.E. and Hernon, P. (2002), "Outcomes assessment: not synonymous with inputs and outputs", *The Journal of Academic Librarianship*, Vol. 28 No. 6, pp. 376-380, doi: 10.1016/S0099-1333(02)00339-7.

Duranti, L. and Blanchette, J.-F. (2004), "The authenticity of electronic records: the InterPARES approach", *paper presented at Archiving 2004*, San Antonio, available at: https://www.ingentaconnect.com/content/ist/ac/2004/00002004/00000001/art00046

Duranti, L., Rogers, C. and Thibodeau, K. (2022), "Authenticity", *Journal of the Archives and Records Association*, Vol. 43 No. 2, pp. 188-203, doi: 10.1080/23257962.2022.2054406.

Elstrøm, G.V. and Junge, J.G. (2014), "Self-assessment of the digital repository at the State and University Library, Denmark: a case study", *Paper Presented at 11th International Conference on Digital Preservation*, Melbourne, available at: https://phaidra.univie.ac.at/detail/o:378128.pdf

Evans, T.N.L. (2021), "Are we winning? Other measurables for digital preservation", *paper presented at 17th International Conference on Digital Preservation*, Beijing, available at: https://hdl.handle.net/11353/10.1424951

Frank, R.D. (2020), "The social construction of risk in digital preservation", *Journal of the Association for Information Science and Technology*, Vol. 71 No. 4, pp. 474-484, doi: 10.1002/asi.24247.

Frank, R.D. and Rothfritz, L. (2023), "Designated community: uncertainty and risk", *Journal of Documentation*, Vol. 79 No. 4, pp. 880-897, doi: 10.1108/JD-07-2022-0161.

Gerke, D.-M., Uude, K. and Kliewe, T. (2023), "Co-creation and societal impact: toward a generic framework for research impact assessment", *Evaluation*, Vol. 29 No. 4, pp. 489-508, doi: 10.1177/13563890231195906.

Giaretta, D. (2011), *Advanced Digital Preservation*, Springer, Berlin, doi: 10.1007/978-3-642-16809-3.

Golub, K., Ziolkowski, P.M. and Zlodi, G. (2022), "Organizing subject access to cultural heritage in Swedish online museums", *Journal of Documentation*, Vol. 78 No. 7, pp. 211-247, doi: 10.1108/JD-05-2021-0094.

Halvarsson, E., Mason, S., Pretlove, L., Gerrard, D., Langley, S. and Mooney, J. (2019), "Could collaborative research between two major libraries help consolidate digital preservation and break the 'project cycle'?", in Myntti, J. and Zoom, J., (Eds) *Digital Preservation in Libraries: Preparing for a Sustainable Future*, ALA, pp. 241-262.

Hart, S. and Amos, H. (2018), "The library assessment capability maturity model: a means of optimizing how libraries measure effectiveness", *Evidence Based Library and Information Practice*, Vol. 13 No. 4, pp. 31-49, doi: 10.18438/eblip29471.

Hedstrom, M. and Lee, C. (2002), "Significant properties of digital objects: definitions, applications, implications", *paper presented at DLM Forum*, Barcelona, available at: https://pdfs.semanticscholar.org/8735/6572cc184bb86a9a27315dc6dec41e8cd0cd.pdf

Hjørland, B. (2023), "Subject (of documents)", Hjørland, B. and Gnoli, C., (Eds) *Encyclopedia of Knowledge Organization*, (3.1 ed.), ISKO, available at: https://www.isko.org/cyclo/subject

Hjørland, B. (2024), "Bibliography", Hjørland, B. and Gnoli, C., (Eds) *Encyclopedia of Knowledge Organization*, ISKO, available at: https://www.isko.org/cyclo/bibliography

Hockx-Yu, H. and Knight, G. (2008), "What to preserve? Significant properties of digital objects", *International Journal of Digital Curation*, Vol. 3 No. 1, pp. 141-153, doi: 10.2218/ijdc.v3i1.49.

Houghton, B. (2017), "Looking back: a university library's self-assessment against ISO 16363:2012", Alexandria, Vol. 27 No. 2, pp. 1-12, doi: 10.1177/0955749017725938.

ISO (2012a), *Space Data and Information Transfer Systems — Audit and Certification of Trustworthy Digital Repositories*, ISO, ISO 16363), available at: https://public.ccsds.org/Pubs/652x0m1.pdf

ISO (2012b), *Space Data and Information Transfer Systems — Open Archival Information System (OAIS) — Reference Model (ISO 14721)*, ISO, available at: https://public.ccsds.org/pubs/650x0m2.pdf

Downloaded from www.emerald.com/jd/article-pdf/81/3/747/9723661/jd-12-2024-0313.pdf by Diponegoro University user on 10 November 2026

ISO (2016), *Information and Documentation — Records Management — Part 1: Concepts and Principles (ISO 15489-1)*, ISO, Geneva, available at: https://www.iso.org/standard/62542.html

Jenatabadi, H.S. (2015), "An overview of organizational performance index: definitions and measurements", *Social Science Research Network*, doi: 10.2139/ssrn.2599439.

Kastenhofer, J. (2015), "The logic of archival authenticity: ISO 15489 and the varieties of forgeries in archives", *Archives and Manuscripts*, Vol. 43 No. 3, pp. 166-180, doi: 10.1080/01576895.2015.1074085.

Kim, D.J. (2012), "An investigation of the effect of online consumer trust on expectation, satisfaction, and post-expectation", *Information Systems and e-Business Management*, Vol. 10 No. 2, pp. 219-240, doi: 10.1007/s10257-010-0136-2.

Knight, G. and Pennock, M. (2009), "Data without meaning: establishing the signifcant properties of digital research", *The International Journal of Digital Curation*, Vol. 4 No. 1, p. 16, doi: 10.2218/ijdc.v4i1.86.

Kuhlthau, C.C. (2017), "Information search process (ISP) model", in McDonald, J.D. and Levine-Clark, M., (Eds) *Encyclopedia of Library and Information Sciences*, (4th ed.), CRC Press, pp. 2232-2238, doi: 10.1081/E-ELIS4.

Lasswell, H.D. (1948), "The structure and function of communication in society", in Bryson, L. (Ed.), *The Communication Of Ideas: A Series Of Addresses*, Jewish Theological Seminary of America, pp. 215-228.

Lee, C.A. and Tibbo, H. (2007), "Digital curation and trusted repositories: steps towards success", *Journal of Digital Information*, Vol. 8 No. 2, available at: https://journals.tdl.org/jodi/index.php/jodi/article/view/229/183

Leonardi, P.M. (2010), "Digital materiality? How artifacts without matter, matter", *First Monday*, Vol. 15 No. 6, doi: 10.5210/fm.v15i6.3036.

MacNeil, H. and Mak, B. (2007), "Constructions of authenticity", *Library Trends*, Vol. 56 No. 1, pp. 26-52, doi: 10.1353/lib.2007.0054.

Manuti, A., Traversa, R. and Mininni, G. (2012), "The dynamics of sense making: a diatextual approach to the intersubjectivity of discourse", *Text & Talk*, Vol. 32 No. 1, pp. 39-61, doi: 10.1515/text-2012-0003.

Mascolo, M.F. (2016), "Beyond objectivity and subjectivity: the intersubjective foundations of psychological science", *Integrative Psychological and Behavioral Science*, Vol. 50 No. 4, pp. 543-554, doi: 10.1007/s12124-016-9357-3.

Matthews, J.R. (2015), *Library Assessment in Higher Education*, 2nd ed., ABC-CLIO, Santa Barbara.

McMeekin, S. and Currie, A. (2022), "Ain't no mountain high enough: developing a new comptency framework for digital preservation", *Paper Presented at 18th International Conference on Digital Preservation*, Glasgow.

Medina-Smith, A. (2021), "A self-audit of the NIST public data repository using the CoreTrustSeal trustworthy data repositories requirements", doi: 10.6028/NIST.IR.8341.

Mingers, J. and Willcocks, L. (2023), "Semiotic foundations: information", (Ed.) *Meaning and Embodied Cognition*, Palgrave Macmillan, Cham, doi: 10.1007/978-3-031-34299-8.

Mois, M., Klas, C.-P. and Hemmje, M.L. (2009), "Digital preservation as communication with the future", *paper presented at 16th International Conference on Digital Signal Processing*, Santorini, pp. 1-8, doi: 10.1109/icdsp.2009.5201104, available at: http://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=5201104

Moles, N. (2021), "Open government data (OGD): challenging the concept of a 'designated community'", *Records Management Journal*, Vol. 31 No. 1, pp. 18-33, doi: 10.1108/RMJ-10-2019-0065.

Moore, R. (2008), "Towards a theory of digital preservation", *International Journal of Digital Curation*, Vol. 3 No. 1, pp. 63-75, doi: 10.2218/ijdc.v3i1.42.

Morris, C. (1964), *Signification and Significance: A Study on the Relations of Signs and Values*, MIT Press, Cambridge, MA.

Downloaded from www.emerald.com/jd/article-pdf/81/3/747/9723661/jd-12-2024-0313.pdf by Diponegoro University user on 10 November 2026

Newcomer, K. and Brass, C.T. (2016), "Forging a strategic and comprehensive approach to evaluation within public and nonprofit organizations: integrating measurement and analytics within evaluation", *American Journal of Evaluation*, Vol. 37 No. 1, pp. 80-99, doi: 10.1177/1098214014567144.

Nöth, W. (1990), *Handbook of Semiotics*, First paperback edition, Indiana University Press, Bloomington, IN.

Nuyts, J. (2006), "Modality: overview and linguistic issues", in Frawley, W., (Ed.) *The Expression of Modality*, De Gruyter Mouton, pp. 1-26, doi: 10.1515/9783110197570.

Nyitray, K.J. and Reijerkerk, D. (2021), "Digital repository legacies: a case study in assessing organizational trustworthiness", *Journal of Library Administration*, Vol. 61 No. 7, pp. 793-812, doi: 10.1080/01930826.2021.1972729.

Owens, T. (2018), *The Theory and Craft of Digital Preservation*, Johns Hopkins University Press, Baltimore.

Peterson, A., Robertson, H. and Szydlowski, N. (2016), "Do you count? The revitalization of a national preservation Statistics survey", *Library Resources and Technical Services*, Vol. 60 No. 1, p. 14, doi: 10.5860/lrts.60n1.38.

Phillips, M., Bailey, J., Goethals, N. and Owens, T. (2013), "The NDSA levels of digital preservation: an explanation and uses", available at: http://www.digitalpreservation.gov/ndsa/working_groups/documents/NDSA_Levels_Archiving_2013.pdf

Pinnick, J. (2017), "Exploring digital preservation requirements: a case study from the National Geoscience Data Centre (NGDC)", *Records Management Journal*, Vol. 27 No. 2, pp. 175-191, doi: 10.1108/RMJ-04-2017-0009.

Peirce, C.S. (1932), *Collected Papers of Charles Sanders Peirce – Volumes I and II: Principles of Philosophy and Elements of Logic*, Harvard University Press, Cambridge.

Poole, A.H. (2016), "The conceptual landscape of digital curation", *Journal of Documentation*, Vol. 72 No. 5, pp. 961-986, doi: 10.1108/JD-10-2015-0123.

Recker, J., Dillo, I., L'Hours, H., Kleemota, M., Crabtree, J., Downs, R.R., Hugo, W., Jenkins, R., Lin, D., Mokrane, M., Trilsbeek, P., von Stein, I. and Callaghan, L. (2020), "An introduction to the CoreTrustSeal trustworthy data repositories requirements, 2020-2022", doi: 10.5281/zenodo.3638211.

Rieger, O.Y., Schonfeld, R.C. and Sweeney, L. (2022), "The effectiveness and durability of digital preservation and curation systems", doi: 10.18665/sr.316990.

Robertson, M., Gattschalk, T. and Wheeler, J. (2024), "The CARL library impact framework: a logic model approach to impact assessment for research libraries", *Partnership: the Canadian Journal of Library and Information Practice and Research*, Vol. 18 No. 2, pp. 1-16, doi: 10.21083/partnership.v18i2.7612.

Rodrigues, E., Ferreira, M., Carvalho, J., Principle, P., Faria, L., Silva, H. and Moreira, J. (2014), "Large scale repository auditing to ISO 16363", *Paper Presented at Open Repositories*, Helsinki, available at: https://urn.fi/URN:NBN:fi-fe2014070432247

Ross, S. (2006), "Approaching digital preservation holistically", in Tough, A. and Moss, M., (Eds) *Record Keeping in a Hybrid Environment: Managing the Creation, Use, Preservation, and Disposal of Unpublished Information Objects in Context*, Chandos, pp. 115-153, doi: 10.1016/B978-1-84334-142-0.50006-2.

Rothenberg, J. (2000), "Preserving authentic digital information", (Eds) *Authenticity in a Digital Environment*, Council on Library and Information Resources, pp. 51-68, available at: https://www.clir.org/wp-content/uploads/sites/6/pub92.pdf

SAA (2020), "Dictionary of archives terminology", *Society of American Archivists*, available at: https://dictionary.archivists.org/(accessed February 2, 2021).

SAA (2024), "Digital preservation", *Society of American Archivists*, available at: https://dictionary.archivists.org/entry/digital-preservation.html (accessed 7 January 2024).

Saussure, F.D. (1983), "Course in general linguistics", in Harris, R.T., Bally, C. and Seschehaye, A., (Eds) *Open Court, La Salle, (Cours de linguistique générale, Payot, Paris, 1916)*.

Downloaded from www.emerald.com/jd/article-pdf/81/3/747/9723661/jd-12-2024-0313.pdf by Diponegoro University user on 10 November 2026

Savolainen, R. (2015), "The interplay of affective and cognitive factors in information seeking and use: comparing Kuhlthau's and Nahl's models", *Journal of Documentation*, Vol. 71 No. 1, pp. 175-197, doi: 10.1108/JD-10-2013-0134.

Schüller-Zwierlein, A. (2015), "Why preserve? An analysis of preservation discourses", *Preservation, Digital Technology and Culture*, Vol. 44 No. 3, pp. 98-122, doi: 10.1515/pdtc-2015-0020.

Shala, L. and Shala, A. (2016), "File formats – characterization and validation", *IFAC-PapersOnLine*, Vol. 49 No. 29, pp. 253-258, doi: 10.1016/j.ifacol.2016.11.06277.

Slade, S., Pearson, D. and Coufal, L. (2014), "A digital preservation environment maturity matrix for NSLA libraries", *Paper Presented at 11th International Conference on Digital Preservation*, Melbourne.

Strodl, S., Becker, C., Neumayer, R. and Rauber, A. (2007), "How to choose a digital preservation strategy: evaluating a preservation planning procedure", *Paper Presented at 7th ACM/IEEE-CS Joint Conference on Digital Libraries*, Vancouver, pp. 29-38, doi: 10.1145/1255175.1255181.

Sundqvist, A. (2021), "Things that work – meditations on materiality in archival discourse", *Journal of Contemporary Archival Studies*, Vol. 8 No. 7, available at: https://elischolar.library.yale.edu/jcas/vol8/iss1/7

Thibodeau, K. (2002), "Overview of technolgical approaches to digital preservation and challenges in coming years", *Paper Presented at The State of Digital Preservation: An International Perspective*, Washington, available at: https://www.clir.org/pubs/reports/pub107/pub107.pdf

Thorpe, C. and Howlett, A. (2020), "Applied and conceptual approaches to evidence-based practice in research and academic libraries", *LIBER Quarterly*, Vol. 30 No. 1, pp. 1-17, doi: 10.18352/lq.10320.

Tiwari, A. and Madalli, D.P. (2021), "Maturity models in LIS study and practice", *Library & Information Science Research*, Vol. 43 No. 1, 101069, doi: 10.1016/j.lisr.2020.101069.

Traczyk, T. (2017), "Requirements for digital preservation", in Traczyk, T., Ogryczak, W., Pałka, P. and Śliwiński, T., (Eds) *Digital Preservation: Putting it to Work*, Springer, pp. 3-14.

UNESCO (2003), *Guidelines for the Preservation of Digital Heritage*, Paris, available at: http://unesdoc.unesco.org/images/0013/001300/130071e.pdf

UNESCO (2019), "Concept of digital preservation", available at: https://en.unesco.org/themes/information-preservation/digital-heritage/concept-digital-preservation (accessed 7 January 2024).

van Veenendaal, R., Lucker, P.C.M. and Sijtsma, C.D. (2018), "Significant significant properties", *Paper Presented at 15th International Conference on Digital Preservation*, Boston, available at: https://openpreservation.org/wp-content/uploads/2018/10/Significant-Significant-Properties.pdf

Wang, X. and Yu, H. (2005), "How to break MD5 and other hash functions", *Lecture Notes in Computer Science*, Vol. 3494, pp. 19-35, Springer, doi: 10.1007/11426639_2.

Waters, D. and Garrett, J. (1996), *Preserving Digital Information: Report of the Task Force on Archiving of Digital Information*, CLIR, Washington, available at: http://www.clir.org/pubs/reports/reports/pub63watersgarrett.pdf

Wilson, A. (2005), "Performance model and process for preserving digital records for long-term access", *Paper Presented At Archiving 2005*, Washington.

Wilson, T.C. (2017), "Rethinking digital preservation: definitions, models, and requirements", *Digital Library Perspectives*, Vol. 33 No. 2, pp. 128-136, doi: 10.1108/DLP-08-2016-0029.

Wyatt, P. (2021), "Demystifying PDF through a machine-readable definition", *Paper Presented at 42nd IEEE Symposium on Security and Privacy Workshops*, available at: https://github.com/gangtan/LangSec-papers-and-slides/raw/main/langsec21/papers/Wyatt_LangSec21.pdf

Yeo, G. (2010), "Nothing is the same as something else': significant properties and notions of identify and originality", *Archival Science*, Vol. 10 No. 2, pp. 85-116, doi: 10.1007/s10502-010-9119-9.

Corresponding author
Stephen Abrams can be contacted at: stephen_abrams@harvard.edu

For instructions on how to order reprints of this article, please visit our website:
www.emeraldgrouppublishing.com/licensing/reprints.htm
Or contact us for further details: **permissions@emeraldinsight.com**

Downloaded from www.emerald.com/jd/article-pdf/81/3/747/9723661/jd-12-2024-0313.pdf by Diponegoro University user on 10 November 2026